

ZAALI Mohamed

AI Agent Developer

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EDUCATION

ENSAM Rabat - State Engineer Student in Data Science, AI & Digital Health

Sep 2023–Present

PROFESSIONAL EXPERIENCE

Syntax - Casablanca - AI Agent Developer Intern

Feb 2026 - Present

Building an extensible MCP server connecting AI agents to Microsoft 365 via Microsoft Graph API.

- Reduced agent configuration complexity by **architecting a preset system** that ships context-aware tool sets.
- Built a **5-step pipeline** that auto-generates typed MCP tools from Microsoft's OpenAPI spec.
- Enabled any M365 service to be onboarded with **zero code changes** by designing a JSON-driven extensibility architecture.

Al Barid Bank - Rabat - Full Stack Developer Intern

Jul 2025 - Sep 2025

Design and development of a platform for analyzing POS, SoftPOS, and e-commerce transactions.

- Accelerated transaction data access by building a **secure REST API** serving aggregated POS, SoftPOS, and e-commerce metrics.
- Empowered analysts to explore trends by delivering a **responsive dashboard** with interactive charts and drill-down navigation.
- Ensured long-term maintainability by designing the full **UML software architecture** before implementation, reducing rework risk.

Ministry of Youth, Culture and Communication - Rabat - Data Science Intern

Jul 2024 - Sep 2024

Development of a predictive analytics system for cultural heritage preservation.

- Enabled proactive preservation decisions by building a **predictive model** forecasting cultural heritage degradation risk.
- Made insights accessible to non-technical experts by integrating model predictions into an interactive **Flask web application**.
- Ensured production availability by deploying the application on an **Ubuntu server** with Gunicorn and Nginx.

PROJECTS & ACCOMPLISHMENTS

- AL-MUQARRIR - Automatic Transcription & Meeting Minutes Generation:** Web platform for **real-time transcription**, speaker **diarization**, and automatic generation of **meeting minutes** with **semantic analysis**. | *Technologies:* Whisper (OpenAI) PyAnnote Gemini API Flask Angular SQLite
- Cura - Disease & Medication Tracking:** Web application for treatment management with AI-powered **drug interaction detection** and generation of patient-friendly alerts. | *Technologies:* PyTorch Gemini API Flask SQLAlchemy Bootstrap SQLite
- LipReader - Lip-Reading Application:** Application converting lip movements into text. | *Technologies:* TensorFlow Streamlit
- NER - Named Entity Recognition Models:** Developed an application capable of identifying named entities. | *Technologies:* Python Streamlit Keras SpaCy Stanza
- Real-Time Network Intrusion Detection System:** System capturing and analyzing network packets in real-time. | *Technologies:* Python Kafka PyShark Pandas Scikit-learn Joblib
- Anxiety Level Prediction:** Platform for assessing and predicting anxiety levels based on psychological, demographic, and behavioral information. | *Technologies:* Python Scikit-learn Flask Bootstrap Joblib
- MySQL-to-Hadoop Pipeline:** Designed and implemented a data migration pipeline from a MySQL database to HDFS via Sqoop, with analysis using Hive. | *Technologies:* Apache Sqoop Hadoop HDFS Apache Hive MySQL Docker Linux Shell
- BigMart Sales Analysis - Dashboard in Google Looker:** Retail analytics dashboard providing strategic insights on sales and inventory optimization. | *Technologies:* Google Looker Python [View Dashboard](#)

TECHNICAL SKILLS

Programming Languages: Python Java scala

AI/ML: scikit-learn, PyTorch, TensorFlow, NLP

Databases: MySQL Oracle MongoDB SQLite PostgreSQL

Big Data: Hadoop Hive Sqoop Kafka Spark

Data Integration: Talend Data Warehousing ETL

Visualization & BI: Power BI Google Looker

Web Technologies & Frameworks: HTML CSS JavaScript Bootstrap Angular Streamlit Flask Spring Boot

EXTRACURRICULAR ACTIVITIES

Learning Experience Architect - Fatal Error Club ENSAM Rabat

SEP 2024 - JUN 2025

- Led workshops in data science, machine learning, and software development; co-created innovative programs.
- Optimized training outcomes through continuous evaluation and refinement.

CERTIFICATIONS

- **CCNA: Introduction to Networks**
- **Data Structures and Algorithms in Python**

LANGUAGES

- **Introduction to Data**
 - **Python Programming Fundamentals**
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- **English:** Fluent
 - **French:** Fluent
 - **Arabic:** Native